Sleep Stage Scoring with Polysomnography (PSG)

Nils Wagner [1] Martin Gaens [1]

[1]Reykjavik University

Introduction to Sleep Stages

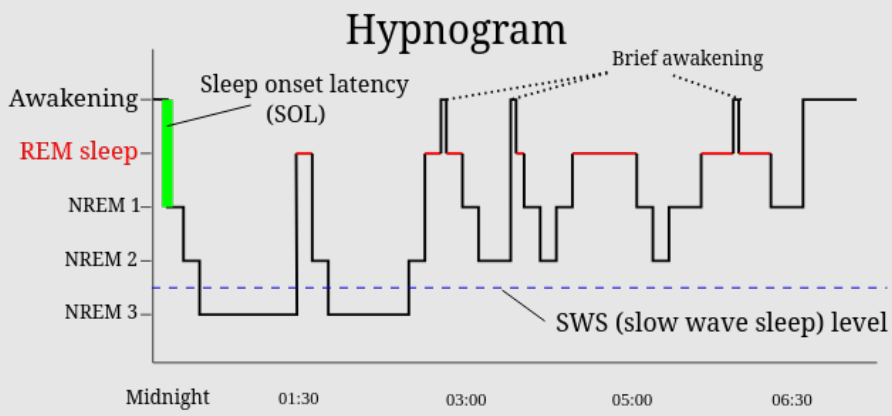


Figure 1. Hypnogram of healthy adult

- Sleep is a biological process essential for brain function, cardiovascular health, immune regulation, and emotional stability
- Two main types: **Non-REM sleep** (stages N1, N2, N3) and **REM sleep**
- Each stage has distinct physiological characteristics measurable through EEG, EOG, and EMG

Sleep Stage Characteristics and Functions

- **REM (Dream Sleep)**: $20 - 25\%$ of total sleep; essential for emotional regulation and procedural memory consolidation
- **N1 (Light Transitional)**: $< 5\%$ of total sleep; transition between wakefulness and sleep
- **N2 (Stable Light Sleep)**: largest proportion of total sleep; supports sensory processing and memory consolidation
- **N3 (Deep Sleep)**: critical for physical restoration, immune function, and declarative memory consolidation; reduced with aging

Sleep Scoring

- According to guidelines in American Academy of Sleep Medicine (AASM) manual
- Recording divided into **30-second epochs**, each classified by **sleep stage** based on EEG, EOG and EMG signals
- **Respiratory events** (apneas, hypopneas) scored according to airflow, effort and oxygen desaturation criteria
- Arousals, movement-related events, and snoring also annotated

Signals from Frontal EEG in Self-applied PSG [1]

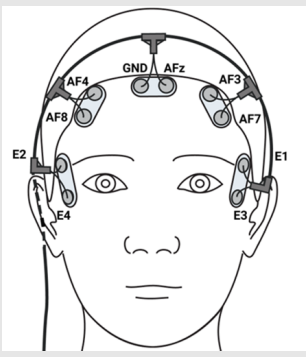


Figure 2. Placement of electrodes in self-applied PSG

- *Eye movement (EOG)*: referenced against AFz and cross referenced
- *Brain activity (EEG)*: referenced against average of E3 and E4
- *Muscle tone (EMG)*: EMG temporalis derived from E2-E4 (left) and E1-E3 (right) signal
- No dedicated chin EMG (like in Type 1 PSG)

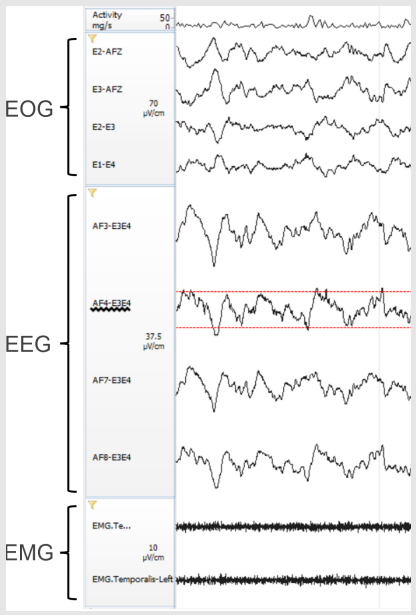
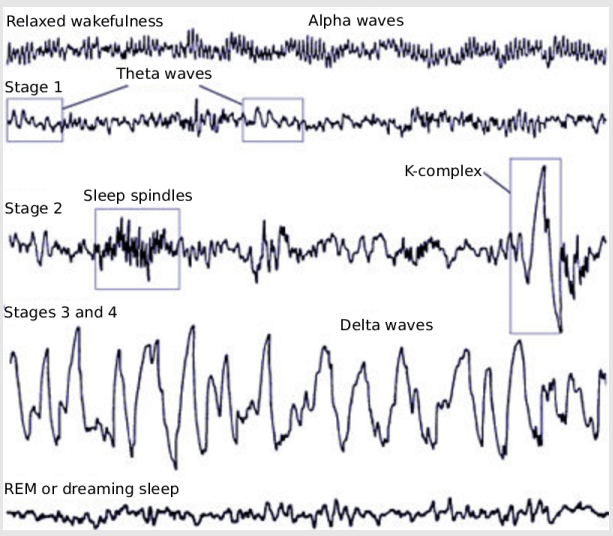


Figure 3. Channels for sleep stage scoring

Characteristics of sleep stages



- **Wake**: alpha and beta waves, high muscle tone
- **REM**: rapid eye movements, muscle atonia (lowest muscle tone)
- **N1**: theta waves, low amplitude and high frequency, brief transitional stage
- **N2**: sleep spindles and K-complexes, stable light sleep
- **N3**: high-amplitude, low-frequency synchronized waves, deep restorative sleep

Figure 4. Characteristic brain wave patterns in different sleep stages [3]

Beta waves (>13 Hz), alpha waves (8-13 Hz), theta waves (4-7 Hz), and delta waves (<4 Hz) are distinguished by their frequency

Polysomnography (PSG)

- Sleep recording/research tool measuring multiple biosignals during sleep
- Records brain activity (EEG), eye movements (EOG), muscle tone (EMG), and respiratory parameters
- Gold standard method for evaluating sleep and sleep disorders
- Used in clinical practice and sleep research
- Types: traditional in-lab PSG (Type 1), self-applied PSG at home (Type 2)

Set-up of self-applied polysomnography

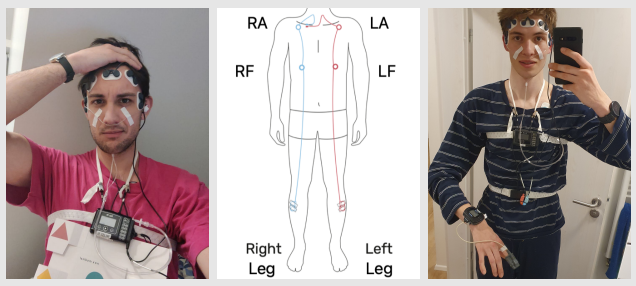


Figure 5. Setup of self-applied polysomnography

- EEG, EOG and facial EMG from electrodes in the face
- ECG and leg EMG from body cables
- Thorax and abdomen belt, nasal pressure
- Pulse oximeter for oxygen saturation
- Audio and snoring signal, position, activity

Difficulty of Sleep Scoring

Different sleep scorers may disagree when classifying certain sleep stages, particularly N1 and N3, as found in [4]. This is also evident in Figure 6.

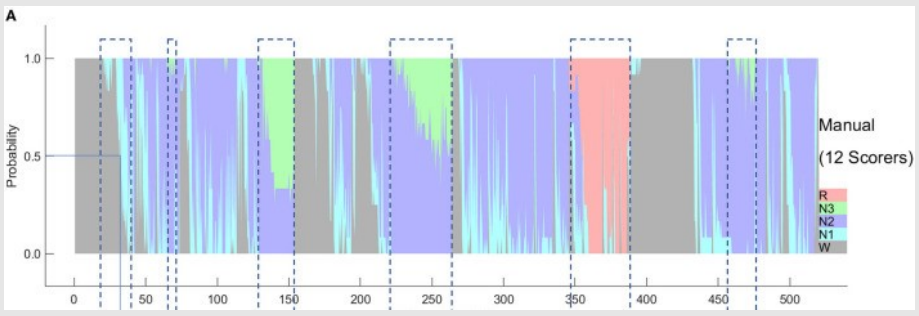


Figure 6. Confidence plot of scorings of 12 sleep scorers [2]

References

[1] Nox Medical Support. *Nox SAS Signals and Impedance Measurements*. https://support.noxmedical.com/hc/en-us/articles/10535298667804-Nox-SAS-Signals-and-Impedance-Measurements. Last updated August 20, 2024. 2024.

[2] P. Anderer, M. Ross, A. Cerny, R. Vasko, E. Shaw, and P. Fonseca. "Overview of the hypnodensity approach to scoring sleep for polysomnography and home sleep testing". In: *Frontiers in Sleep* 2 (Apr. 2023). ISSN: 2813-2890. DOI: 10.3389/frsle.2023.1163477. URL: http://dx.doi.org/10.3389/frsle.2023.1163477.

[3] S. Hitziger. "Modeling the variability of electrical activity in the brain". In: (Apr. 2015).

[4] R. S. Rosenberg and S. Van Hout. "The American Academy of Sleep Medicine Inter-scorer Reliability Program: Sleep Stage Scoring". In: *Journal of Clinical Sleep Medicine* 9.1 (Jan. 2013), pp. 81–87. ISSN: 1550-9397. DOI: 10.5664/jcsm.2350. URL: http://dx.doi.org/10.5664/jcsm.2350.

 nils25@ru.is, martin25@ru.is